

Using Terraform to create an Ubuntu EC2 instance with a security group and a PEM key, and configure user data to unzip and host a static website using an HTTP server

Terraform code:

```
#Declare provider as AWS
provider "aws" {
  access_key = "aws-access-key"
  secret_key = "aws-secret-key "
  region     = "us-east-2"
}

# Generate Key Pair Locally
resource "tls_private_key" "my_key" {
  algorithm = "RSA"
  rsa_bits  = 2048
}

resource "aws_key_pair" "deployer" {
  key_name   = "my-key"
  public_key = tls_private_key.my_key.public_key_openssh
}

# Save Private Key Locally
resource "local_file" "private_key" {
  content = tls_private_key.my_key.private_key_pem
  filename = "${path.module}/my-key.pem"
}

# Security Group for EC2
resource "aws_security_group" "web_sg" {
  name        = "web-sg"
  description = "Allow inbound HTTP and SSH"

  # Allow SSH
  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
  }
}
```

Terraform Commands:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\devops\Terraform> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated
with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.web will be created
+ resource "aws_instance" "web" {
  + ami                        = "ami-0cb91c7de36eed2cb"
  + arn                       = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone          = (known after apply)
  + cpu_core_count             = (known after apply)
  + cpu_threads_per_core       = (known after apply)
  + disable_api_termination    = (known after apply)
```

Resources created in AWS:

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Find Instance by attribute or tag (case-sensitive)

All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ..
☒	UbuntuWebSe...	i-037e2021ff1351979	Running	t2.micro	2/2 checks passed	View alarms +	us-east-2b	ec2-3-129-65-197.us-e...	3.129.65.197

Security group, inbound and outbound rules:

i-037e2021ff1351979 (UbuntuWebServer)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role

Owner ID

Launch time

sg-053f66067d3a6ba32 (web-sg)

354918371559

Thu Feb 27 2025 21:37:51 GMT+0530 (India Standard Time)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-02555e0275808ad0c	22	TCP	0.0.0.0/0	web-sg	-
-	sgr-08c7227d2e84ab6de	80	TCP	0.0.0.0/0	web-sg	-

▼ Outbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
-	sgr-0787a8250b1590220	All	All	0.0.0.0/0	web-sg	-

Static website running inside the webserver public IP:

